

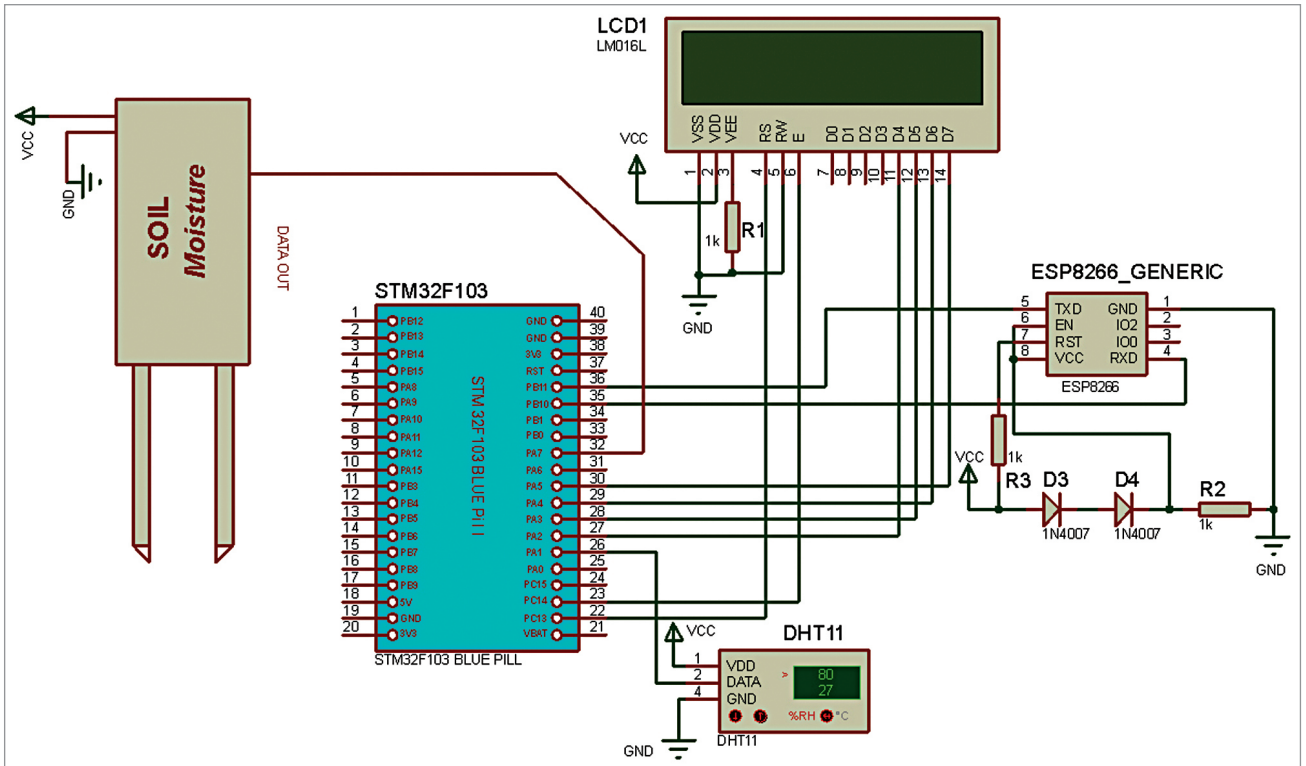


Fig. 3: The main circuit diagram

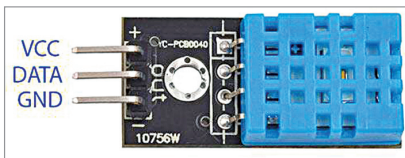


Fig. 4: Pin configuration of DHT11

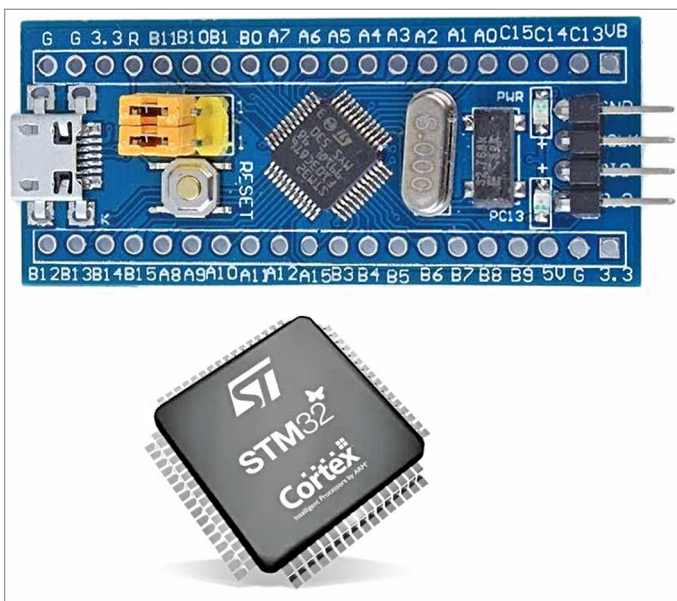


Fig. 5: STM32F103C8T6 (Blue Pill)

of DHT11 are shown in Fig. 4 and the STM32F103C8T6 (Blue Pill) is shown in Fig. 5.

Voltage regulator for STM32 can be replaced by 5V battery or 5V DC adaptor. The STM microcontroller is

used for getting the sensor data and displaying it on LCD and also sending the data to the Wi-Fi board. The ESP receives the data and uploads it to ThingSpeak cloud. But before that we need to set up the ThingSpeak and get the API token to incorporate

it in the code.

For uploading the data to ThingSpeak IoT platform and registration refer Fig. 6 through Fig. 12 and follow the steps given below.

- First create an account on ThingSpeak.com, sign in and click on Get Started.
- Go to the Channels menu and click on New Channel option on the same page.
- You will see a form for creating the channel, where you may fill in Name and Description as per your choice. Then fill Temperature, Humidity, and Soil Level in Field 1, Field 2, and Field 3 and tick the check boxes for all three fields. Also tick the check box for Make Public option below in the form, and finally save the channel. Now the new channel has been created.
- Next, click on API Keys tab and save the Write and Read API keys. Here we are using only Write key, which needs to be copied in `char *api_key` in the code.